OntoCoDec: Near-Complete Type-Valid Information Extraction via Compiled Ontology Constraints

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Abstract

Large language models (LLMs) applied to information extraction (IE) produce systematic ontology constraint violations—15.4% of extracted triples violate domain/range type constraints on SciERC and 18.7% on DocRED. We show that these violations concentrate on relation types with tight type constraints (e.g., Compare: 42.4%) rather than scaling with ontology size, and that prompt-level intervention, while reducing overall violations by 62%, remains unreliable at the per-relation level, with one relation type exhibiting *increased* violations.

We present OntoCoDec, a framework that automatically compiles IE ontology schemas into two composable constraint layers for LLM decoding: (1) a *grammar layer* enforcing JSON structure and type/relation vocabulary via context-free grammars, and (2) a *semantic layer* enforcing cross-field domain/range constraints through a stateful entity registry with dynamic logit masking. The two layers compose via bitwise AND with an enabling dependency: grammar ensures vocabulary validity, enabling the semantic layer to look up and enforce domain/range rules.

Evaluated on SciERC (7 relations) and DocRED (96 relations) with Qwen3-8B and Llama-3.1-8B, OntoCoDec reduces domain/range violations by 94% on SciERC (15.4%→0.93%) and 44.7–90.2% on DocRED, while eliminating entity-type and relation-vocabulary violations entirely. F1 impact is neutral on SciERC (p=0.58) and significantly positive on DocRED (+0.50pp relation F1, p<0.001). We provide a decidability taxonomy classifying constraint types by their amenability to autoregressive enforcement and identify cross-field consistency as an inherent limitation of per-relation independent enforcement.

1 Introduction

Information extraction (IE) systems built on large language models achieve impressive zero-shot performance by framing extraction as structured generation. However, when an ontology defines type constraints on relations—e.g., that USED-FOR requires a METHOD head and a TASK tail—LLMs routinely violate these constraints in their output.

We conduct the first systematic study of ontology constraint violations in LLM-based IE. Our violation landscape analysis across two datasets (SciERC with 7 relations, DocRED with 96) reveals two key findings: (1) domain/range violations are non-trivial (15.4% and 18.7% respectively) and concentrate on relation types with tight constraints rather than scaling with ontology size; (2) schema-informed prompting reduces violations by 62% overall but is unreliable at the per-relation level—Evaluate-for violations *increase* by 5.8 percentage points even when constraints are explicitly described in the prompt.

These findings motivate OntoCoDec, a framework that compiles ontology schemas into decoder constraints operating at two complementary layers. The **grammar layer** uses context-free grammars (via XGrammar) to enforce JSON structure, entity type vocabulary, and relation name vocabulary—constraints that are decidable from the token prefix alone. The **semantic layer** maintains a stateful entity registry during generation and applies dynamic logit masking to enforce domain/range constraints—which require cross-field state but remain prefix-decidable given the registry.

The two layers exhibit an *enabling dependency*: grammar ensures every generated relation name belongs to the valid vocabulary, which is a prerequisite for the semantic layer to look up the corresponding domain/range rules. Without grammar, the semantic layer encounters out-of-vocabulary relation names and cannot enforce constraints (semantic-only achieves 24% DR reduction vs. full system 94%).

We classify constraints into a decidability taxonomy: (a) purely local constraints (entity type $\in$ vocabulary), enforceable by grammar alone; (b) prefix-decidable with state (domain/range compatibility), requiring the semantic layer's entity registry; and (c) truly non-local constraints (forward coreference), which are out of scope for forward-only autoregressive decoding.

Our contributions:

1. A systematic violation landscape study establishing that LLM-based IE produces non-trivial, tightness-driven ontology violations, and that prompt-level intervention is insufficient (§3).
2. A composable two-layer constraint framework with principled decidability taxonomy, compiling ontology schemas into grammar and semantic decoder constraints (§4).
3. Comprehensive experiments on 2 datasets $\times$ 2 models showing 94% DR elimination on SciERC at zero F1 cost, with honest analysis of the cross-field consistency trade-off (§5).
4. A cross-model generalization study demonstrating zero-code-modification transfer across model families, with a dual-factor analysis explaining DR reduction variance (§5).

2 Background

IE as Structured Generation. Modern LLM-based IE frames extraction as generating JSON-structured output containing entity lists and relation triples. Each triple (h, r, t) associates a head entity h with type τ_h, a relation r, and a tail entity t with type τ_t. An ontology schema $\mathcal{O}$ defines the valid entity types $\mathcal{T}$, relation types $\mathcal{R}$, and for each $r \in \mathcal{R}$, the domain $\text{dom}(r) \subseteq \mathcal{T}$ and range $\text{ran}(r) \subseteq \mathcal{T}$.

Constraint Violations. We define four violation types:

- **Entity type (ET)**: $\tau_h \notin \mathcal{T}$ or $\tau_t \notin \mathcal{T}$
- **Relation vocabulary (RV)**: $r \notin \mathcal{R}$
- **Domain/range (DR)**: $\tau_h \notin \text{dom}(r)$ or $\tau_t \notin \text{ran}(r)$
- **Cross-field consistency (CF)**: same entity assigned different types across relations

Grammar-Constrained Decoding. Recent work on grammar-constrained decoding (e.g., XGrammar) enforces context-free grammar constraints during autoregressive generation by masking logits at each decoding step. This guarantees structural validity (e.g., valid JSON) but cannot enforce cross-field semantic constraints that depend on previously generated content.

Constraint Tightness. We define the *constraint tightness* τ_r of a relation r as:

$$\bar{\tau}_r = 1 - \frac{|\text{dom}(r)| \times |\text{ran}(r)|}{|\mathcal{T}|^2} \tag{1}$$

Higher tightness means fewer valid type combinations, predicting higher violation rates.

3 Violation Landscape Study

We conduct a systematic study of ontology constraint violations in zero-shot LLM-based IE using Qwen3-8B on SciERC (551 abstracts, 7 relations) and DocRED (998 documents, 96 relations).

Table 1: Per-relation DR violation rates on SciERC (Qwen3-8B, minimal prompt). $\bar{\tau}$: constraint tightness.

Relation	**#Triples**	**DR (%)**	$\bar{\tau}$
Compare	—	42.4	high
Feature-of	—	33.5	high
Part-of	—	25.0	med
Evaluate-for	—	24.1	med
Hyponym-of	—	14.5	med
Used-for	—	0.8	low
Conjunction	—	0.0	low
Overall	**1,724**	**15.4**	—

Table 2: Schema-informed prompt effect on DR violation rate by relation (SciERC, Qwen3-8B). All differences significant by McNemar's test.

Relation	**DR_{min} (%)**	**DR_{schema} (%)**	**Δpp**	**Dir.**
Feature-of	33.5	11.3	−22.3	↓
Part-of	25.0	2.5	−22.5	↓
Compare	42.4	27.6	−14.8	↓
Evaluate-for	24.1	29.9	+5.8	↑
Overall	**15.4**	**5.78**	**−62.4%**	↓

3.1 SciERC Violation Profile

Table 1 presents per-relation domain/range violation rates under a minimal prompt (relation list only, no constraint descriptions). The overall DR violation rate is 15.4% (265 of 1,724 triples), but the distribution is highly non-uniform: Compare exhibits 42.4% violations while Used-for shows only 0.8%.

The 56× gap between Compare (42.4%) and Used-for (0.8%) directly tracks constraint tightness: Compare accepts only 2–3 valid (head_type, tail_type) pairs, while Used-for accepts 5 of 6 entity types as heads. This supports our central claim that **violation rate is driven by constraint tightness, not ontology size**.

Full violation profile: ET = 0.4%, RV = 1.39%, DR = 15.37%, CF = 0.08%. JSON parse rate is 100%. Entity F1 = 31.01%, Relation F1 = 8.23% (expected for zero-shot).

3.2 Prompt-Level Intervention

Table 2 shows the effect of schema-informed prompting, where each relation's domain/range constraints are explicitly described in the prompt.

Schema-informed prompting achieves a 62.4% overall reduction (15.4%→5.78%, 1,661 triples), but per-relation results are unreliable. Critically, Evaluate-for violations *increase* by 5.8pp. This is not noise but a systematic **prior-constraint conflict**: the model's distributional prior ("Method X evaluates for Task Y" is semantically plausible) conflicts with the schema constraint, destabilizing type assignment. Three relations retain >10% violation rates (Compare 27.6%, Evaluate-for 29.9%, Feature-of 11.3%), demonstrating that prompt-level intervention cannot provide per-instance compliance guarantees.

3.3 Cross-Dataset: DocRED

We extend the violation landscape study to DocRED (998 documents, 96 Wikidata property relations). Table 3 summarizes the results.

DocRED's 18.7% DR is comparable to SciERC's 15.4% despite 14× greater ontology complexity, confirming that violation rates are driven by per-relation constraint tightness rather than ontology

Table 3: DocRED violation landscape (Qwen3-8B). Top: minimal prompt; Bottom: schema-informed prompt.

Prompt	**ET (%)**	**RV (%)**	**DR (%)**	**CF (%)**
Minimal	7.5	28.9	18.7	0.08
Schema-informed	0.61	18.4	12.8	0.15
Reduction	−91.9%	−36.3%	−31.3%	—

scale. The long-tail distribution is pronounced: P35 exhibits 100% DR (but few instances) while P31 (instance_of) shows 60.4% DR with 322 violations.

Schema-informed prompting achieves only 31.3% DR reduction on DocRED versus 62.4% on SciERC, demonstrating diminishing returns with ontology complexity. With 96 relations, constraint descriptions create information overload that exceeds the model's working memory capacity.

RV = 28.9% is a DocRED-specific challenge: 96 Wikidata property IDs exceed the model's zero-shot vocabulary memory. This RV–DR orthogonality (high RV, comparable DR) becomes central to the cross-model analysis in §5.5.

4 OntoCoDec Framework

OntoCoDec automatically compiles an IE ontology schema $\mathcal{O} = (\mathcal{T}, \mathcal{R}, \text{dom}, \text{ran})$ into two composable constraint layers applied during autoregressive decoding.

4.1 Grammar Layer

The grammar layer compiles $\mathcal{O}$ into a context-free grammar (CFG) enforced via XGrammar during token generation. This CFG guarantees:

- Valid JSON structure (brackets, quotes, commas)
- Entity types restricted to $\mathcal{T}$: `entity_type` $\in$ {"*Method*", "*Task*", ... }
- Relation names restricted to $\mathcal{R}$: `relation` $\in$ {"*Used-for*", "*Compare*", ... }

These are purely local constraints (decidability class (a))—each token's validity depends only on the grammar state and the token itself, not on previously generated semantic content.

4.2 Semantic Layer

The semantic layer implements a `TypeConstraintProcessor` that maintains an *entity registry* $\mathcal{E}$ mapping entity mentions to their assigned types during generation. When the decoder reaches a type-constrained position (e.g., the head entity type of a relation triple):

1. Look up the current relation r from the partially generated output.
2. Retrieve $\text{dom}(r)$ from the compiled constraint table.
3. Compute a logit mask M_{semantic} allowing only tokens corresponding to types in $\text{dom}(r)$.
4. Apply M_{semantic} via element-wise multiplication with the logit vector.

This is a prefix-decidable constraint with state (decidability class (b)): the constraint depends on the current relation name (already generated) and the ontology rules (compiled ahead of time), requiring the entity registry as state.

4.3 Composable Constraint Application

At each decoding step, the grammar layer produces mask M_{grammar} and the semantic layer produces mask M_{semantic}. The final mask is:

$$M_{\text{final}} = M_{\text{grammar}} \wedge M_{\text{semantic}} \tag{2}$$

Enabling Dependency. Grammar is a prerequisite for semantic: the semantic layer looks up domain/range rules by relation name, which requires the relation name to be in $\mathcal{R}$. Without grammar ensuring vocabulary validity, the semantic layer encounters unknown relation names and cannot enforce constraints.

Graceful Degradation. When $M_{\text{grammar}} \cap M_{\text{semantic}} = \emptyset$ (high constraint tightness creates empty intersection), the system falls back to grammar-only control, maintaining structural validity as a guaranteed floor. We quantify this via a *Constraint Coverage* metric: the fraction of type-token positions where semantic masking is active (i.e., non-fallback).

4.4 Decidability Taxonomy

We classify IE ontology constraints by their amenability to autoregressive enforcement:

- **Class (a) – Purely local**: entity type $\in \mathcal{T}$, relation $\in \mathcal{R}$. Enforceable by CFG (grammar layer).
- **Class (b) – Prefix-decidable with state**: $\tau_h \in \text{dom}(r)$, $\tau_t \in \text{ran}(r)$. Requires entity registry (semantic layer).
- **Class (c) – Truly non-local**: forward coreference consistency (same entity, same type across all mentions). Requires future context; out of scope for forward-only decoding.

Cross-field consistency (CF) violations arise from class (c) constraints: ensuring that an entity receives the same type in every relation it participates in requires global coordination that autoregressive decoding cannot provide without multiple passes.

4.5 Decoding Strategies

We implement two decoding strategies:

- **S1 (Single-pass)**: Both layers active throughout generation. Primary configuration.
- **S2 (Two-pass)**: Pass 1 extracts entities with grammar-only constraints; Pass 2 generates relation triples with an input-dependent grammar compiled from Pass 1 entities plus full semantic constraints. Positioned as an architectural ablation.

5 Experiments

5.1 Setup

Datasets. SciERC (551 abstracts, 7 relations, 6 entity types) and DocRED (998 documents, 96 relations). Both provide ontology schemas with domain/range constraints.

Models. Qwen3-8B (primary) and Llama-3.1-8B (cross-model validation). Both evaluated in zero-shot setting with no task-specific training.

Table 4: SciERC 4-way ablation (Qwen3-8B). DR: domain/range violation rate. Bold: primary configuration.

Config	ET (%)	RV (%)	DR (%)	CF (%)	E-F1 (%)	R-F1 (%)
Unconstrained	0.40	1.39	15.37	0.08	31.01	8.23
Grammar-only	0.00	0.00	14.84	0.08	30.98	8.23
Semantic-only	0.40	1.51	11.65	2.71	31.01	8.15
Full S1	**0.00**	**0.00**	**0.93**	**9.90**	**30.98**	**8.36**

Table 5: S1 vs S2 on SciERC (Qwen3-8B). S2 positioned as architectural ablation.

Strategy	#Triples	DR (%)	CF (%)	R-F1 (%)	E-F1 (%)
S1 (single-pass)	1,729	0.93	9.90	8.36	30.98
S2 (two-pass)	1,644	0.97	8.03	8.03	31.90

Evaluation. Violation rates (ET, RV, DR, CF) measured by an automated constraint checker. F1 scores computed using standard entity and relation matching. Statistical significance assessed via 10,000-iteration bootstrap resampling with permutation p-values.

Baselines. (1) Unconstrained: standard generation with no constraints. (2) Schema-informed prompt: explicit constraint descriptions in the prompt (§3.2). (3) Post-hoc filtering: remove triples with ET or DR violations from unconstrained output.

5.2 SciERC Ablation Study

Table 4 presents the 4-way ablation on SciERC, isolating each layer's contribution.

Grammar-only eliminates ET and RV entirely ($\rightarrow$0%) but barely affects DR (14.84%, -3.4% relative). Grammar ensures valid vocabulary but cannot enforce cross-field type constraints. CF remains unchanged (0.08%), confirming grammar does not introduce CF.

Semantic-only reduces DR by 24% (15.37%$\rightarrow$11.65%) but leaves ET and RV untouched—these are grammar's responsibility. The limited DR reduction occurs because the semantic layer encounters out-of-vocabulary relation names (RV 1.51%) and cannot look up their domain/range rules.

Full S1 achieves 94% DR reduction (15.37%$\rightarrow$0.93%, 16 residual violations). The 70pp gap between semantic-only (-24%) and full system (-94%) quantifies the enabling dependency: grammar's vocabulary enforcement is a prerequisite for effective semantic constraint enforcement. If the layers were independently additive, the predicted full-system DR would be $\sim$11.3%, far above the observed 0.93%.

F1 Impact. Entity F1 is unchanged (-0.03pp, p=0.53), relation F1 shows a non-significant increase ($+0.13$pp, p=0.58). Constraint enforcement is F1-neutral on SciERC.

Residual DR. The 16 residual violations concentrate on Compare and Evaluate-for (high tightness), corresponding to fallback events where $M_{\text{semantic}} \cap M_{\text{grammar}} = \emptyset$. Fallback frequency: $\sim$10.7 events per document, indicating $\sim$85–90% semantic masking coverage.

Table 6: DocRED 4-way ablation (Qwen3-8B). †: $p<0.001$ vs. unconstrained (bootstrap).

Config	ET (%)	RV (%)	DR (%)	CF (%)	E-F1 (%)	R-F1 (%)	#OOV
Unconstr.	2.48	29.66	18.46	0.10	58.15	7.74	5,189
Gram.-only	0.00	0.03	26.17	0.07	59.38	8.31	5
Sem.-only	4.22	29.87	13.84	1.05	57.81	7.70	—
Full S1	**0.00**	**0.03**	**10.21**	**2.57**	**59.37**†	**8.23**†	**5**

Table 7: Fallback frequency and residual DR: SciERC vs DocRED.

Dataset	Fallback/doc	$\mathbf{DR_{residual}}$ (%)	$\bar{\tau}$
SciERC	∼10.7	0.93	moderate
DocRED	∼57	10.21	0.842

5.3 S1 vs S2 Architectural Comparison

Table 5 compares the single-pass (S1) and two-pass (S2) strategies.

S2 does not solve CF: the 1.87pp reduction (9.90%→8.03%) is not statistically significant. S2 also extracts fewer triples (1,644 vs. 1,729) and achieves lower relation F1 (8.03% vs. 8.36%). Subtype analysis reveals that S2 reduces entity-vs-relation type drift (EVR) by 27% but increases within-relation inconsistencies ($10 \rightarrow 18$ entities). S1 is the recommended configuration; S2 demonstrates framework flexibility in supporting multiple decoding strategies.

5.4 DocRED Ablation Study

Table 6 presents the DocRED ablation.

Latent DR Reveal. The most counterintuitive finding: grammar-only *increases* DR from 18.46% to 26.17% (+7.71pp). Grammar forces out-of-vocabulary relation names onto valid vocabulary, revealing latent DR violations previously hidden in unrecognizable triples. This "latent DR reveal" mechanism is central to understanding DocRED's constraint dynamics.

Full S1 reduces DR by 44.7% (18.46%→10.21%). While weaker than SciERC's 94%, this is explained by the dual-factor model (§5.5): high RV baseline (29.66%) means grammar reveals many latent DR violations, and high ontology tightness ($\bar{\tau}$=0.842) creates more fallback events (∼57/doc vs. ∼10.7/doc on SciERC).

F1 Impact. Both F1 metrics improve significantly: entity F1 +1.23pp ($p<0.001$, 95% CI [+0.94, +1.55]), relation F1 +0.50pp ($p<0.001$, 95% CI [+0.23, +0.81]). The mechanism is grammar eliminating 5,189 OOV relation names (all counted as false positives) and non-standard entity types, improving precision. Grammar-only entity F1 (59.38%) ≈ full S1 (59.37%), confirming that the grammar layer is the sole source of F1 improvement; the semantic layer contributes zero additional F1.

CF Trade-off. DocRED CF = 2.57%, far below SciERC's 9.90%. DocRED's entity type system (PER, LOC, ORG, etc.) has clearer semantic boundaries than SciERC's (Generic, OtherScientificTerm, etc.), resulting in more consistent cross-relation type assignment.

Table 7 confirms the direct relationship between fallback frequency and residual DR: DocRED's 5.3× higher fallback rate (57 vs. 10.7 per document) corresponds to 11× higher residual DR.

Table 8: Complete 2×2 cross-model × cross-dataset results. PF: parse failure rate (%).

Model	Dataset	Config	ET (%)	RV (%)	DR (%)	CF (%)	R-F1 (%)	E-F1 (%)	PF (%)	DR Δ
Qwen3-8B	SciERC	Unconstr.	0.40	1.39	15.37	0.08	8.23	31.01	0.0	—
		Full S1	0.00	0.00	0.93	9.90	8.36	30.98	0.0	−94.0
Qwen3-8B	DocRED	Unconstr.	2.48	29.66	18.46	0.10	7.74	58.15	1.2	—
		Full S1	0.00	0.03	10.21	2.57	8.23	59.37	1.1	−44.7
Llama-3.1-8B	SciERC	Unconstr.	0.00	0.55	13.34	0.47	2.68	23.56	0.2	—
		Full S1	0.00	0.00	0.08	6.56	2.54	23.49	0.0	−99.4
Llama-3.1-8B	DocRED	Unconstr.	0.13	16.03	26.07	0.17	5.98	55.65	6.8	—
		Full S1	0.00	0.00	2.55	12.02	6.27	55.01	9.1	−90.2

Table 9: Dual-factor analysis: constraint tightness ($\bar{\tau}$) × RV baseline predict DR reduction.

Model	Dataset	$\mathbf{RV_{base}}$ (%)	$\mathbf{DR_{unc}}$ (%)	$\mathbf{DR_{S1}}$ (%)	ΔDR (%)
Llama	SciERC	0.55	13.34	0.08	−99.4
Qwen	SciERC	1.39	15.37	0.93	−94.0
Llama	DocRED	16.03	26.07	2.55	−90.2
Qwen	DocRED	29.66	18.46	10.21	−44.7

5.5 Cross-Model Generalization

We evaluate OntoCoDec on Llama-3.1-8B with zero code modifications—same prompt, same pipeline, different model weights. Table 8 presents the complete 2×2 (model × dataset) matrix.

OntoCoDec generalizes across model families without code modification. DR reductions span a wide range (44.7%–99.4%), with ET and RV eliminated entirely (→0%) in all four cells.

RV–DR Orthogonality. Llama shows lower RV (16.03% vs. 29.66% on DocRED) but *higher* DR (26.07% vs. 18.46%). Vocabulary compliance and semantic compliance are orthogonal failure modes—knowing the correct relation name does not imply knowing the correct type constraints. This motivates the two-layer design: any model may be strong on one dimension and weak on the other.

Dual-Factor Analysis. Table 9 reveals that DR reduction effectiveness is predicted by two factors: (1) ontology constraint tightness $\bar{\tau}$ and (2) model RV baseline.

The DR reduction ordering ($99.4\% > 94.0\% > 90.2\% > 44.7\%$) perfectly mirrors the RV baseline ordering ($0.55\% < 1.39\% < 16.03\% < 29.66\%$). Higher RV baseline → more OOV relations → grammar reveals more latent DR violations → more fallback events → higher residual DR. This dual-factor model (tightness × RV baseline) explains the DR reduction variance across all four experimental cells.

F1 Behavior. Llama shows F1-neutral behavior on SciERC (−0.14pp relation F1) and a slight improvement on DocRED (+0.29pp), consistent with the Qwen pattern. Entity F1 on Llama-DocRED shows a minor decrease (−0.64pp), possibly related to increased parse failures (6.8%→9.1%) under constrained decoding.

5.6 Post-Hoc Filtering Baseline

Table 10 compares OntoCoDec with post-hoc filtering (removing triples with ET or DR violations from unconstrained output).

Table 10: Multi-dimensional comparison: OntoCoDec vs. post-hoc filtering (Qwen3-8B).

	Method	R-F1 (%)	Recall (%)	Reject. (%)	#TP
SciERC	Unconstr.	8.23	11.40	—	111
	OntoCoDec	8.36	11.60	0.0	—
	Post-hoc	8.63	10.68	16.8	104
DocRED	Unconstr.	7.74	9.46	—	1,137
	OntoCoDec	8.23	10.02	0.0	—
	Post-hoc	9.61	8.40	48.1	1,010

Table 11: Bootstrap CI for F1 impact (10,000 iterations, Qwen3-8B).

Dataset	Metric	Δ(pp)	95% CI	p-value
SciERC	Rel F1	$+0.13$	—	0.58
SciERC	Ent F1	-0.03	—	0.53
DocRED	Rel F1	$+0.50$	[$+0.23$, $+0.81$]	<0.001
DocRED	Ent F1	$+1.23$	[$+0.94$, $+1.55$]	<0.001

Post-hoc filtering achieves higher F1 than OntoCoDec (SciERC: 8.63 vs. 8.36; DocRED: 9.61 vs. 8.23) by aggressively removing false-positive triples and boosting precision. However, multi-dimensional comparison reveals fundamental trade-offs:

Recall Preservation. OntoCoDec *increases* recall ($+0.20$pp SciERC, $+0.56$pp DocRED) through probability mass redistribution to legal tokens. Post-hoc filtering *decreases* recall (-0.72pp, -1.06pp) through information deletion.

Information Coverage. Post-hoc filtering discards 48.1% of DocRED triples and 16.8% of SciERC triples. On DocRED, 127 correctly extracted triples (true positives) are lost because they co-occur with entity type violations—these are valid extractions that OntoCoDec preserves by enforcing legal types at generation time. This is the core "prevent vs. detect" argument.

ET Residual. Post-hoc filtering removes triples but does not modify the entity list, leaving residual ET violations: 2.47% on DocRED vs. 0% for OntoCoDec.

5.7 Statistical Significance

Table 11 reports bootstrap confidence intervals (10,000 iterations) for F1 differences between unconstrained and full S1.

Constraint enforcement is F1-neutral on SciERC (both $p > 0.5$) and significantly positive on DocRED ($p < 0.001$ for both metrics). The dataset-dependent pattern is mechanistically explained: DocRED's high RV baseline (29.66%) means grammar eliminates many OOV false positives, improving precision and F1. SciERC's low RV baseline (1.39%) provides minimal such opportunity.

5.8 Cross-Field Consistency Analysis

The elevated CF rate under full S1 (SciERC: 9.90%, DocRED: 2.57%, Llama-DocRED: 12.02%) is an inherent consequence of per-relation independent type enforcement. Ablation confirms that CF is entirely introduced by the semantic layer: grammar-only CF = 0.08% (unchanged from unconstrained), while full S1 CF = 9.90%.

Subtype Analysis. On SciERC, >90% of CF violations are entity-vs-relation type drift (EVR): the same entity receives different types in the entity list versus relation triples. The top conflict patterns—Generic↔OtherScientificTerm (60 instances), Material↔OtherScientificTerm (28), Material↔Method (20)—involve SciERC entity types with inherently ambiguous semantic boundaries.

Severity Distinction. DR violations produce structurally invalid triples (the downstream knowledge graph cannot accept them). CF violations leave each individual triple type-valid; only cross-triple consistency is broken. This is a soft preference rather than a hard constraint.

Cross-Dataset Pattern. DocRED CF (2.57%) is far lower than SciERC (9.90%), because DocRED's entity types (PER, LOC, ORG) have clearer semantic boundaries than SciERC's abstract scientific types. However, Llama-DocRED CF (12.02%) is elevated, likely related to Llama's weaker format stability on DocRED (9.1% parse failures).

6 Related Work

LLM-Based Information Extraction. Zero-shot and few-shot IE with LLMs has shown promise across entity, relation, and event extraction tasks. However, the systematic characterization of ontology constraint violations in LLM-generated extractions has not been previously studied.

Constrained Decoding. Grammar-constrained decoding enforces structural constraints (JSON validity, type safety) during generation. Recent work such as XGrammar compiles structural constraints into efficient token masks. OntoCoDec extends this paradigm to semantic constraints that require cross-field state, going beyond what context-free grammars can express.

Schema-Guided Generation. Schema-informed prompting and in-context learning provide constraint information to LLMs. Our violation landscape study shows that this approach is insufficient for per-instance compliance, motivating decoding-time enforcement.

Post-Hoc Validation. An alternative to generation-time constraints is post-hoc filtering or correction. We provide an empirical comparison showing that post-hoc filtering trades information coverage for precision, losing correctly extracted triples as collateral damage.

Type-Constrained Knowledge Graph Completion. Type constraint enforcement has been studied in knowledge graph embedding, but these approaches operate on discrete symbol prediction rather than autoregressive text generation, making them inapplicable to LLM-based IE.

7 Limitations and Future Work

Cross-Field Consistency. CF violations (6.5–12.0% across configurations) are an inherent limitation of per-relation independent enforcement. A global entity-type consensus mechanism (enforcing consistent types before relation-specific enforcement) is a promising direction but requires non-trivial extensions to the autoregressive framework.

Model and Dataset Scope. We validate on 2 models $\times$ 2 datasets. While the zero-code-modification transfer to Llama demonstrates generalization, additional model families (particularly larger models and instruction-tuned variants) would strengthen the claim.

Zero-Shot Setting. F1 values are low ($\sim$8% relation F1) as expected in zero-shot extraction. The constraint enforcement method is evaluated as an orthogonal technique; integration with few-shot or fine-tuned extractors is future work.

Fallback Degradation. On high-tightness ontologies, the semantic$\cap$grammar empty-intersection fallback occurs frequently ($\sim$57 times/doc on DocRED), degrading to grammar-only control. Improved fallback strategies (e.g., relaxed constraints, constraint prioritization) could reduce residual DR.

8 Conclusion

We presented OntoCoDec, a framework for near-complete type-valid information extraction through compiled ontology constraints. Our violation landscape study establishes that LLM-based IE produces systematic, tightness-driven ontology violations that prompt-level intervention cannot reliably eliminate. OntoCoDec addresses this through two composable constraint layers—grammar for vocabulary validity, semantic for domain/range enforcement—with an enabling dependency and graceful degradation.

Across 2 models $\times$ 2 datasets, OntoCoDec eliminates entity-type and relation-vocabulary violations entirely, reduces domain/range violations by 44.7–99.4%, and maintains or improves extraction F1. The dual-factor analysis (tightness $\times$ RV baseline) provides a principled explanation for the variance in DR reduction across experimental conditions.

We identify cross-field consistency as a fundamental limitation of autoregressive per-relation enforcement and provide a decidability taxonomy classifying constraint types by their amenability to forward-only decoding. This taxonomy offers theoretical guidance for future work on increasingly expressive constraint frameworks for structured generation.

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